

Eric Shaw

📍 Dallas, TX

✉ engineofsky@gmail.com

☎ +1 863 703 2416

🌐 in/eric-shaw-0262a512/

Visionary Result-Driven Senior AI/ML Engineer with over 10 years of proven expertise designing and deploying state-of-the-art machine learning solutions that drive innovation and deliver measurable business impact. Specialized in Natural Language Processing (NLP), Large Language Models (LLMs), Computer Vision, Generative AI, and Conversational AI, with deep proficiency in MLOps and LLMOps practices for scalable, production-grade deployment. Adept at translating complex theoretical models into high-impact applications across industries, bridging research and enterprise-grade execution. Recognized for leading cross-functional, globally distributed teams in fast-paced, agile environments - fostering collaboration between engineering, product, and business stakeholders. Demonstrates a strategic mindset with strong technical leadership, driving AI adoption from ideation through to end-to-end delivery. Passionate about building intelligent systems that are not only innovative but aligned with long-term organizational goals, operational excellence, and ethical AI practices.

PROFESSIONAL EXPERIENCE

Amira Learning

Senior AI/ML Engineer

California, United States

Nov 2024 – Nov 2025

- **Architected a multi-agent conversational AI framework** for real-time English tutoring across text and voice channels, applying ReAct and Chain-of-Thought reasoning with Model Context Protocol (MCP) to ensure consistent state handling and memory alignment across agents.
- Implemented **hybrid Knowledge Retrieval** using Pinecone vector search and **GraphRAG** (Neo4j) to dynamically surface cultural nuances, idioms, and context-aware learning prompts - improving **response accuracy** and **learner comprehension** by 40%.
- Built and optimized a **PEFT fine-tuning pipeline** (QLoRA, Adapters, Instruction Tuning) on GPT-4 / Llama models through **LangGraph**, reducing training cycles by **40%** and **cutting hallucination rate** in language instruction scenarios by 60%.
- Deployed **full MLOps infrastructure** using Terraform + AWS EKS/ECR with Lambda, Step Functions, and API Gateway for scalable conversational tutoring workloads - increasing peak session throughput **3× without quality loss**.
- Developed an **Automated Tutoring Quality Evaluation Suite** using LangSmith + OpenAI Evals to measure learning relevance, tone appropriateness, and cultural sensitivity - enabling continuous A/B iteration on tutor persona behaviors.
- **Designed the multi-modal learning engine** enabling synchronized **audio, text, and visual flashcard learning**, adaptive pronunciation guidance, and real-time learner proficiency scoring integrated with Snowflake learner analytics.

Zazmic Inc

Data Scientist

Texas, United States

Aug 2017 – Nov 2024

- Built advanced clinical NLP pipelines on **Databricks and Snowflake** for processing **EHR** and claims data, including a **BERT-based medical entity recognition system** (**ICD, CPT, medications, symptoms**) using **spaCy** + **TensorFlow** - reduced chart-extraction error rates from 11% to 2%.
- Developed **multi-agent clinical decision-support chat systems** using **LLM-based orchestration** (**LangChain Agents**) and adaptive context-aware healthcare agents, reducing manual clinician review workload by 60% and streamlining care-team workflows.
- Deployed **RLHF pipelines** with **Weights & Biases** on Google Cloud **Vertex AI** to continuously tune clinical NLP models using physician/annotator feedback, cutting false positives in risk-factor and diagnosis extraction by 25%.
- Created a real-time medical literature & **guideline summarization platform** (**GPT, NLTK, PySpark**) that synthesizes research papers, FDA updates, and clinical notes - reduced medical analyst review time by 75%.

- Implemented a patient-insight sentiment and quality-of-care analysis engine (**PyTorch**, **GCP AutoML**), achieving 92% accuracy across 500,000+ patient reviews, enabling more data-driven operational decisions.
- Orchestrated full **MLOps** lifecycle for healthcare AI models using **Kubeflow**, **Terraform**, and **CloudFormation**, accelerating compliant deployments by 30% while maintaining strict integration testing and automated LLM validation for safe, stable releases.
- Built LDA and BERT-based **clinical topic-modeling systems (Gensim)** with graph embeddings to classify 10M+ clinical records annually, improving coding/granularity accuracy by 42% and strengthening downstream analytics for population-health teams.

IBM

Machine Learning Engineer

Texas, United States

Jun 2014 – Jul 2017

- Engineered an automated **OCR** pipeline (**Tesseract**, **OpenCV**, **Keras**) for client onboarding, slicing manual data entry from 100,000 to 30,000 hours annually and streamlining business processes.
- Boosted product engagement by 45% through a **recommendation engine** built with **scikit-learn** and **PyTorch** for e-commerce personalization, driving higher click-throughs and conversions.
- Deployed a **classification system** for healthcare documents (**TensorFlow**, **NLTK**), achieving 97% accuracy on 1,000 daily records and refining data management practices.
- Developed robust time-series forecasting models (**ARIMA**, **LSTM**, **GRU**) to detect risk patterns and predict future trends, enabling proactive action for clients in healthcare and retail.
- Created executive-grade dashboards in Tableau for real-time customer behavior insights, influencing data-driven strategies that realized a 20% surge in campaign engagement and ROI.

EDUCATION

University of Texas at Austin

Master of Science in Computer Science

Austin, TX, United States

2011 – 2014

University of North Texas

Bachelor of Science in Computer Science

Dallas, TX, United States

2007 – 2011

SKILLS

- **Programming & Tools** – Python, SQL, FastAPI, Flask, RESTful APIs, Git, JIRA, VSCode, Jupyter
- **Machine Learning & Deep Learning** – Supervised & Unsupervised & Self Supervised Learning, DNN (CNNs, RNNs, Transformers), Model Optimization & Compression Reinforcement Learning, Multi-modal Learning
- **Natural Language Processing & LLMs** – LLMs (GPT, BERT, T5, Llama, Claude, etc.), Model Context Protocol(MCP), LLM Orchestration, Prompt Engineering, Fine-Tuning, Conversational AI, Dialogue systems, RAG(Retrieval-Augmented Generation), Text Classification, NER, Sentiment Analysis,
- **Computer Vision** – Image Classification & Object Detection, Semantic Segmentation, OCR, Diffusion Models, GANs, VAEs
- **MLOps / LLMOps** – AWS, GCP, Azure, Docker, Kubernetes, Helm, Terraform, Model Deployment & Serving (ONNX, TensorRT, vLLM, TensorRT-LLM, RayServe, Kserve, Triton, TorchServe), CI/CD for ML Pipeline (Github Actions, Jenkins), Model Monitoring (LangSmith, Prometheus, Kafka, Arize), Vector Database (Pinecone, FAISS, Weaviate, Neo4j), LangChain, LangGraph, LlamaIndex
- **Data Engineering** – Data Pipelines(Apache Airflow, Prefect), Spark, BigQuery
- **Soft Skills** – Technical Leadership & Mentorship, Cross-Functional Collaboration, Agile & Remote Team management, Strategic Thinking, Problem Solving, Product-Minded Engineering